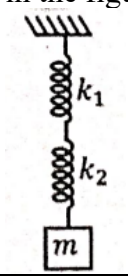
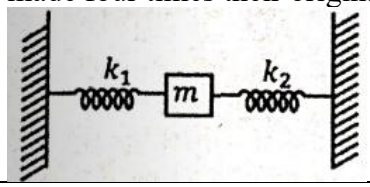
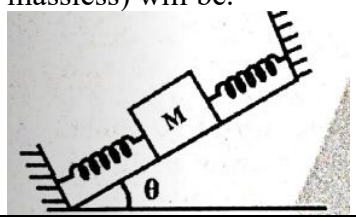
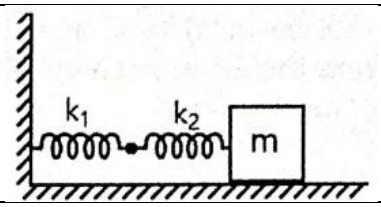
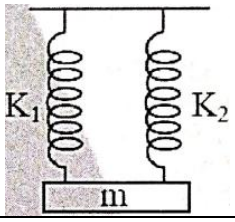
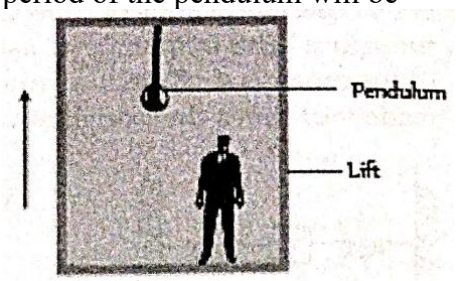
Question	The motion of a particle varies with time according to the relation $y = \alpha \sin \omega t + a\cos\ \omega t$. Then:
Type	multiple choice
Option	The motion is oscillatory but not SHM
Option	The motion is SHM with an amplitude a $\sqrt{2}$
Option	The motion is SHM with an amplitude $\sqrt{2}$
Option	The motion is SHM with an amplitude a
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	The equations of the displacement of two particles making SHM are represented by $y_1 = a\sin(\omega t + \varphi)$and $y_2 = a\cos(\omega t)$ respectively. The phase difference of the velocities of the two particles will be
Type	multiple choice
Option	$\pi / 2 + \varphi$
Option	$-\varphi$
Option	φ
Option	$\varphi - \pi / 2$
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	The displacement of a particle executing SHM is given by y = 0.25 (sin 200t) cm. The maximum speed of the particle is:
Type	multiple choice
Option	200 cm/sec
Option	100 cm/sec
Option	50 cm/sec
Option	0.25 cm/sec
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	A particle undergoes SHM with a time period of 2 seconds. In how much time will it travel from its mean position to a displacement equal to half of its amplitude?
Type	multiple choice
Option	$\frac{1}{2}s$
Option	$\frac{1}{6}s$
Option	$\frac{1}{4}s$
Option	$\frac{1}{3}s$
Answer	2
Solution	
Positive Marks	4
Negative Marks	1

Question	Two simple harmonic motions of angular frequency $100 rad\, s^{-1}$ and 1000 $rad\ s^{-1}$ have the same displacement amplitude. The ratio of their maximum acceleration will be:
Type	multiple choice
Option	$1:10$
Option	$1:10^2$
Option	$1:10^3$
Option	$1:10^4$
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	The maximum velocity of a simple harmonic motion represented by $y = 3\sin\left(100t + \frac{\pi}{6}\right)$ is given by
Type	multiple choice
Option	300
Option	$\frac{3\pi}{6}$
Option	100
Option	$\frac{\pi}{6}$
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	The displacement equation of a particle is $x = 3\sin 2t + 4\cos 2t$. The amplitude and maximum velocity will be respectively
Type	multiple choice
Option	$5,10$
Option	$3,2$
Option	$4,2$
Option	$3,4$
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	The amplitude of a particle executing S.H.M with frequency of 60 Hz is 0.01 m. The maximum value of the acceleration of the particle is
Type	multiple choice
Option	$144\pi^2 m/\sec^2$
Option	$144 m/\sec^2$
Option	$\frac{144}{\pi^2} m/\sec^2$
Option	$228\pi^2 m/\sec^2$
Answer	1
Solution	
Positive Marks	4
Negative Marks	1

Question	What is the maximum acceleration of the particle doing the SHM $y = 2\sin\left[\frac{\pi t}{2} + \phi\right]$ where 2 is in cm
Type	multiple choice
Option	$\frac{\pi}{2} cm/s^2$
Option	$\frac{\pi^2}{2} cm/s^2$
Option	$\frac{\pi}{4} cm/s^2$
Option	$\frac{\pi^4}{4} cm/s^2$
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	Which one of the following equations of motion represents simple harmonic motion? Where k, k_0, k_1 and α are all positive.
Type	multiple choice
Option	Acceleration $= -k_0 x + k_1 x^2 k_0$
Option	Acceleration $= -k(x+a)$
Option	Acceleration $= k(x+a)$
Option	Acceleration $= kx$
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	A mass m is suspended from two springs of spring constant k_1 and k_2 as shown in the figure below. The time period of vertical oscillations of the mass will be. 
Type	multiple choice
Option	$2\pi\sqrt{\left(\frac{k_1+k_2}{m}\right)}$
Option	$2\pi\sqrt{\frac{m}{(k_1+k_2)}}$
Option	$2\pi\sqrt{\frac{m(k_1k_2)}{(k_1+k_2)}}$
Option	$2\pi\sqrt{\frac{m(k_1+k_2)}{(k_1k_2)}}$

Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	Two springs of force constants k1 and k2 are connected to a mass m as shown in the figure. The frequency of oscillation of the mass is f. If both k_1 and k_2 are made four times their original values. 
Type	multiple choice
Option	$2f$
Option	$f/2$
Option	$f/4$
Option	$4f$
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	On a smooth inclined plane, a body of mass M is attached between two springs. The other ends of the springs are fixed to firm supports. If each spring has force constant K, the period of oscillation of the body (assuming the spring as massless) will be. 
Type	multiple choice
Option	$2\pi\left(\frac{M}{2K}\right)^{1/2}$
Option	$2\pi\left(\frac{2M}{K}\right)^{1/2}$
Option	$2\pi\frac{Mg\sin\theta}{2K}$
Option	$2\pi\left(\frac{2Mg}{K}\right)^{1/2}$
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	All the surface are smooth and the system, given below is oscillating with an amplitude A. What is the extension of spring having spring constant k_1, when the block is at the extreme Position?

	
Type	multiple choice
Option	$\frac{k_1}{k_1+k_2}A$
Option	$\frac{k_2 A}{k_1+k_2}$
Option	A
Option	$\frac{A}{2}$
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	When a mass is suspended separately by two different springs, in successive order, then the time period of oscillations is t_1 and t_2 respectively. If it is connected by both springs as shown in the figure below, then the time period of oscillation becomes t_0. The correct relation between t_0, t_1 and t_2 is. 
Type	multiple choice
Option	$t_0^2 = t_1^2 + t_2^2$
Option	$t_0^{-2} = t_1^{-2} + t_2^{-2}$
Option	$t_0^{-1} = t_1^{-1} + t_2^{-1}$
Option	$t_0 = t_1 + t_2$
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	A pendulum is hung from the roof of a sufficiently high building and is moving freely to and fro like a simple harmonic oscillator. The acceleration of the bob of the pendulum is $20 m/s^2$ at a distance of 5m from the mean position. The time period of oscillation is
Type	multiple choice
Option	$2\pi s$
Option	πs
Option	$2 s$
Option	$1 s$
Answer	2
Solution	
Positive Marks	4
Negative Marks	1

Question	A simple pendulum has a time period T in vacuum. Its time period when it is completely immersed in a liquid of density one-eight of the density of material of the bob is
Type	multiple choice
Option	$\sqrt{\frac{7}{8}}T$
Option	$\sqrt{\frac{5}{8}}T$
Option	$\sqrt{\frac{3}{8}}T$
Option	$\sqrt{\frac{8}{7}}T$
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	If the length of second's pendulum is decreased by 2%, how many seconds it will lose per day
Type	multiple choice
Option	3927 s
Option	3727 s
Option	3427 s
Option	864 s
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	A simple pendulum is executing simple harmonic motion with a time period T. If the length of the pendulum is increased by 21%, the percentage increase in the time period of the pendulum of increased length is
Type	multiple choice
Option	10%
Option	21%
Option	30%
Option	50%
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	A man measures the period of a simple pendulum inside a stationary lift and finds it to be T s. If the lift accelerates upwards with an acceleration g/4, then the period of the pendulum will be 
Type	multiple choice
Option	T

Option	$\frac{T}{4}$
Option	$\frac{2T}{\sqrt{5}}$
Option	$2T\sqrt{5}$
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	If a body is released into a tunnel dug across the diameter of earth, it executes simple harmonic motion with time period
Type	multiple choice
Option	$T = 2\pi\sqrt{\frac{R_e}{g}}$
Option	$T = 2\pi\sqrt{\frac{2R_e}{g}}$
Option	$T = 2\pi\sqrt{\frac{R_e}{2g}}$
Option	$T = 2$ seconds
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	Choose the correct statement
Type	multiple choice
Option	Time period of a simple pendulum depends on amplitude
Option	Time shown by a spring watch varies with acceleration due to gravity g
Option	In a simple pendulum time period varies linearly with the length of the pendulum
Option	The graph between length of the pendulum and time period is a parabola
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	Two pendulum of lengths 1.44 m and 1 m start to swing together. The number of vibrations after which they will again start swinging together is
Type	multiple choice
Option	4 of larger pendulum
Option	6 of larger pendulum
Option	6 of smaller pendulum
Option	5 of smaller pendulum
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	A simple pendulum, suspended from the ceiling of a stationary van, has time period T. If the van starts moving with a uniform velocity the period of the pendulum will be
Type	multiple choice
Option	Less than T
Option	Equal to 2T

Option	Greater than T
Option	Unchanged
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	The amplitude of an oscillating simple pendulum is 10 cm and its period is 4s. Its speed after 1 s after it passes its equilibrium position, is
Type	multiple choice
Option	Zero
Option	0.57 m/s
Option	0.212 m/s
Option	0.32 m/s
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	A simple pendulum is oscillating with amplitude 'A' and angular frequency 'w'. At displacement 'x' from mean position, the ratio of kinetic energy to potential energy is
Type	multiple choice
Option	$\frac{x^2}{A^2 - x^2}$
Option	$\frac{x^2 - A^2}{x^2}$
Option	$\frac{A^2 - x^2}{x^2}$
Option	$\frac{A - x}{x}$
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	A particle is vibrating in a simple harmonic motion with an amplitude of 4 cm. At what displacement from the equilibrium position, is its energy half potential and half kinetic
Type	multiple choice
Option	$1\ cm$
Option	$\sqrt{2}\ cm$
Option	$3\ cm$
Option	$2\sqrt{2}\ cm$
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	A body performs SHM with an amplitude A. At a distance $A/\sqrt{2}$ from the mean position, the correct relation between KE and PE is
Type	multiple choice

Option	$KE = \frac{PE}{2}$
Option	$KE = \sqrt{2}PE$
Option	$KE = PE$
Option	$KE = \frac{PE}{\sqrt{2}}$
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	When the kinetic energy of a body executing S.H.M. is 1/3 of the potential energy. The displacement of the body is x percent of the amplitude, where x is
Type	multiple choice
Option	33
Option	87
Option	67
Option	50
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	The total energy of the body executing S.H.M. is E. Then the kinetic energy when the displacement is half of the amplitude, is
Type	multiple choice
Option	$\frac{E}{2}$
Option	$\frac{E}{4}$
Option	$\frac{3E}{4}$
Option	$\frac{\sqrt{3}}{4}E$
Answer	3
Solution	
Positive Marks	4
Negative Marks	1